Minimum Extendable Fano Projections and the Five-Cycle Double Cover Conjecture

Atharva Vaidya
AI-assisted research note for independent human verification

July 29, 2026

Resolution status. The five-cycle double cover conjecture remains unresolved. This note proves an exchange theorem, settles the minimum-support cases through size twelve, and gives exact finite censuses. It proves neither the conjecture nor its negation.

Abstract

Hušek and Šámal recently expressed the five-cycle double cover conjecture as the problem of selecting a nowhere-zero $\mathbb{F}_2^3$-flow with a component-parity property. We study the support h of one binary coordinate of such a flow. Call h *extendable* if it occurs as a coordinate of some nowhere-zero $\mathbb{F}_2^3$-flow.

We prove a simultaneous exchange theorem for a minimum-cardinality extendable projection. Write the flow as $f = (h, s)$, with $s : E \to \mathbb{F}_2^2$, and partition h into the four affine value classes $M_c = \{e \in h : s(e) = c\}$. For every c, the projection h is a minimum-cardinality binary cycle containing M_c. Equivalently, every binary cycle C avoiding M_c satisfies

$$|C \cap h| \leq |C - h|.$$

Thus every circuit component of h contains all four affine colours, and every support-shortening cycle meets all four classes. The proof is the addition of the Fano value $(1, c)$ along C.

We first prove, without computation, that every globally minimum extendable projection of size at most five is cleanable. We then give a stronger direct boundary repair. Independently on every component of $G - h$, apply an element of $\mathrm{GL}(2, 2)$ to the low-flow values; on every circuit of h, solve the resulting cyclic vertex equations while allowing zero low values, since the first coordinate is one there. Cleanliness becomes four explicit cut-parity equations per component. An exact affine/dihedral quotient exhausts every support shape, proper four-colour circuit word, and component partition through total support size ten. All 125,178 valid dirty canonical states admit a direct repair. At size eleven the same test first fails: there are 14 failed $5 + 6$ boundary states, in 11 full symmetry orbits. For every failure, however, an explicit boundary certificate makes the six-circuit low-valued and nowhere zero, so deleting it from the first-coordinate support produces an extendable projection of size five. Global minimality excludes all 14. Thus every minimum extendable projection of size at most eleven is cleanable. At size twelve, 13,788,824 dirty boundary states either clean directly or admit a strict circuit deletion; two independent enumerators agree exactly. Thus every minimum extendable projection of size at most twelve is cleanable, and a counterexample to the proposed selection principle must have minimum size at least thirteen.

We perform an exact canonical census of all 41,301 connected simple cubic graphs of order 18. Among the 39,866 bridgeless graphs, exactly 179 are not Tait-colourable, and every minimum-cardinality extendable projection on all 179 is cleanable. A second exact replay uses frozen canonically generated sources: 14,009 cyclically 4-edge-connected non-Tait records through order 28, and a separate 12,892-record hard order-22 source. All 147,539 minimum extendable

projections in those files are cleanable. Completeness of the source populations is inherited from their upstream generation packages; the result on the literal frozen files is checked directly. An additional exact SAT scan covers seven retained order-34 and 7,654 retained order-40 strong-snark rows. All 433,730 minimum projections in those files have size ten and are cleanable.

The universal statement suggested by this computation is recorded as an open conjecture. The missing step is to derive a colour-avoiding strict exchange from a rainbow-odd component, possibly after support-neutral recolouring. We make no claim of a FiveCDC resolution or of priority for the flow formulation.

Contents

1 Status, conventions, and contribution boundary **2**

2 Extendable and clean Fano projections **3**

3 The minimum-projection exchange theorem **4**
- 3.1 Shortest joins and four-colour consequences 5
- 3.2 A line-switch lemma and the size-six/seven boundary 7
- 3.3 Direct component repair through size ten 8
- 3.4 The first direct-repair failures and size eleven 10
- 3.5 Clean or delete at size twelve 11

4 The exact missing step **11**

5 Exact finite census **12**
- 5.1 Canonical order-18 method 12
- 5.2 Results and reproducibility 13
- 5.3 Expanded frozen-source replay 13
- 5.4 Retained strong snarks of orders 34 and 40 14
- 5.5 A certified 130-vertex parameter-separation test 14

6 Discussion **15**

1 Status, conventions, and contribution boundary

A binary cycle of a graph is an Eulerian edge set; it may be empty or disconnected. A k-cycle double cover is an indexed family of k binary cycles in which every edge occurs exactly twice. Empty members may be appended, so "at most five" and "five, allowing empty members" are equivalent. The five-cycle double cover conjecture, attributed to Celmins and Preissmann, asserts that every bridgeless graph has such a cover with at most five members [1, 4].

We work with the cubic flow formulation in [3, Conjecture 3.19]. No general-to-cubic reduction is proved in this note. All theoretical statements below apply to finite loopless cubic multigraphs unless "simple" is stated. Parallel edges are distinct edge objects. A loop would contribute twice to vertex degree; loops are excluded here to keep the cut notation literal. The finite census is restricted to simple graphs.

Hušek and Šámal prove that the cubic FiveCDC problem is equivalent to selecting a nowhere-zero $\mathbb{F}_2^3$-flow with an additional parity condition [3, Theorem 3.16 and Conjecture 3.19]. Their criterion is related to the prescribed-cycle formulation of Hoffmann-Ostenhof [2]. We claim no novelty or priority for either formulation.

The contribution of this note is narrower:

1. a minimum-support exchange theorem, with a complete proof;
2. a human proof through size five and an exact boundary extension through size twelve;
3. four simultaneous shortest-join consequences;
4. a reproducible canonical order-18 census and an expanded exact replay on frozen sources through order 28; and
5. a precise conjectural step that would turn this route into a proof.

The last item is open.

2 Extendable and clean Fano projections

Let $G = (V, E)$ be a loopless cubic graph and put $K = \mathbb{F}_2^2$. Write a three-bit flow as

$$f = (h, s) : E \longrightarrow \mathbb{F}_2 \times K, \tag{1}$$

where $h : E \to \mathbb{F}_2$ and $s : E \to K$ are flows. We identify a binary flow with its support.

Definition 2.1. A binary cycle h, possibly zero, is *extendable* if there is a flow $s : E \to K$ for which $f = (h, s)$ is nowhere zero.

Writing $s = (p, q)$, extendability has the following elementary form:

$$E - h \subseteq p \cup q, \tag{2}$$

where p and q are binary cycles. Indeed, an edge of $E - h$ has first coordinate zero and therefore needs a nonzero low coordinate.

Fix an extension $f = (h, s)$. For $c \in K$, put

$$M_c = \{e \in h : s(e) = c\}. \tag{3}$$

Proposition 2.2. *Every M_c is a matching, and the four sets M_c partition h.*

Proof. The partition statement is immediate. If two edges incident with a cubic vertex both had flow value $(1, c)$, their sum would be zero and flow conservation would force the third incident edge to have value zero, contrary to nowhere-zeroness. □

Let W be a component of the spanning subgraph $(V, E - h)$. Every edge of $\delta(W)$ belongs to h. If

$$n_c(W) = |\delta(W) \cap M_c| \pmod 2,$$

then cut conservation for the first coordinate and the two low coordinates gives

$$\sum_{c \in K} n_c(W) = 0, \qquad \sum_{c \in K} n_c(W)c = 0. \tag{4}$$

The 3×4 binary matrix in (4) has one-dimensional kernel, generated by the all-ones vector. Hence the four parities $n_c(W)$ are equal.

Definition 2.3. An extension is *clean* if $n_c(W) = 0$ for every component W of $(V, E - h)$, for one, and hence every, $c \in K$. A projection h is *cleanable* if it has a clean extension. A component with all four parities equal to one is *rainbow-odd*.

If $s = (p, q)$, the class M_{11} on $\delta(W)$ is $\delta(W) \cap p \cap q$. Thus cleanability is exactly the existence of binary cycles p, q satisfying (2) and

$$|\delta(W) \cap p \cap q| \equiv 0 \pmod 2 \tag{5}$$

for every component W of $(V, E - h)$. By [3, Theorem 3.16], existence of such an h is equivalent to the cubic FiveCDC assertion. In the Tait-colourable case one may take $h = \varnothing$ and a nowhere-zero K-flow.

3 The minimum-projection exchange theorem

Assume now that G is not Tait-colourable, so an extendable projection cannot be empty. Choose h of minimum cardinality among all extendable projections, and fix any extension $f = (h, s)$.

Theorem 3.1 (minimum-projection exchange)**.** *For every $c \in K$ and every binary cycle C with $C \cap M_c = \varnothing$,*

$$|C \cap h| \leq |C - h|. \tag{6}$$

Equivalently, h is a minimum-cardinality binary cycle containing M_c, simultaneously for all four $c \in K$.

Proof. Fix $c \in K$, put $t = (1, c)$, and add t to the flow value on every edge of C:

$$f'(e) = \begin{cases} f(e) + t, & e \in C, \\ f(e), & e \notin C. \end{cases} \tag{7}$$

Since C is a binary cycle, the added function is a flow. The only edges that could become zero in (7) are those whose old value was t, and these are exactly the edges of M_c. The hypothesis $C \cap M_c = \varnothing$ therefore makes f' nowhere zero.

The first coordinate of f' is $h \triangle C$. This is another extendable projection, so minimality gives

$$|h| \leq |h \triangle C| = |h| + |C - h| - |C \cap h|.$$

This is (6).

For the equivalent statement, rebase the low coordinates by replacing s with the flow $s + ch$. Its exact zero set is M_c: outside h, the low value was already nonzero, while on h it vanishes exactly when $s(e) = c$.

Let h' be any binary cycle containing M_c. The flow $(h', s + ch)$ is nowhere zero. On an edge outside h', its low value is nonzero because every zero lies in $M_c \subseteq h'$; on an edge of h', its first coordinate is one. Thus h' is extendable, and $|h| \leq |h'|$.

Conversely, if $C \cap M_c = \varnothing$, then $h \triangle C$ contains M_c. Applying the minimum-containing property to $h \triangle C$ recovers (6). $\square$

3.1 Shortest joins and four-colour consequences

Let T_c be the set of endpoints of M_c. Removing M_c from a binary cycle containing M_c produces a T_c-join in $G - M_c$, and adjoining M_c reverses the correspondence. Therefore Theorem 3.1 has the following form.

Corollary 3.2. *For every $c \in K$, the edge set*

$$J_c = h - M_c$$

is a shortest T_c-join in $G - M_c$.

This is not the already-refuted principle that a cardinality-minimum exact-zero matching must extend to a FiveCDC certificate. Here the objective is the cardinality of the whole binary cycle $M_c \cup J_c$, and the same projection is optimal after all four affine rebases.

Corollary 3.3 (flow-resistance bound). *Let $r_f(G)$ be the minimum number of zero edges in an $\mathbb{F}_2^2$-flow on G. Every extendable projection satisfies*

$$|h| \geq 4r_f(G). \tag{8}$$

Proof. For each c, the rebased low flow $s + ch$ has exact zero set M_c. Hence $|M_c| \geq r_f(G)$. The four M_c's partition h. □

Corollary 3.4 (four-colour circuits). *Every circuit component of h contains at least one edge of each of the four sets M_c.*

Proof. If a circuit component D omitted M_c, then D would be a binary cycle avoiding M_c, but $|D \cap h| = |D| > |D - h| = 0$, contrary to (6). □

Theorem 3.5 (minimum projections through size five). *Let G be a connected bridgeless loopless cubic graph; parallel edges are allowed. If G is not Tait-colourable and its minimum extendable projection has size at most five, then every minimum extendable projection is cleanable. If G is Tait-colourable, the zero projection is a clean minimum.*

Proof. The Tait case follows by taking a nowhere-zero K-flow as the low coordinate: its zero affine class is empty, so the equal cut parities in (4) are all even. Assume henceforth that G is non-Tait, choose a minimum extendable h, and fix an arbitrary extension $f = (h, s)$.

By Corollary 3.4, every circuit component of h meets all four M_c. Since G is loopless, $|h| \leq 5$ therefore forces h to be one ordinary circuit D of length $n = 4$ or 5; a parallel two-circuit cannot meet four classes. Write

$$v_0, e_0, v_1, e_1, \ldots, v_{n-1}, e_{n-1}, v_0, \qquad e_i = v_i v_{i+1},$$

cyclically, and put $c_i = s(e_i)$. Adjacent c_i's are distinct: otherwise the third edge at their common vertex would have full flow value zero.

Let $K_0 = G - h$. Every component of K_0 contains a vertex of D, by connectedness. Suppose the extension is dirty. For a dirty component W, membership along D toggles oddly on each of the four affine colour classes.

If $n = 4$, the four colours occur once each, so membership toggles on every e_i. The components of K_0 are exactly the two alternating vertex pairs. Indeed, separating either pair further would create a component whose cut sees exactly two affine colours oddly, contrary to (4).

If $n = 5$, rotate and rename the colours so their cyclic word is A, B, A, C, D. Put

$$z_i = 1[v_i \in W], \qquad y_i = z_i + z_{i+1}.$$

Dirtiness says

$$y_1 = y_3 = y_4 = 1, \qquad y_0 + y_2 = 1.$$

Up to complement, the solutions are $W = \{v_2, v_4\}$ and $W = \{v_1, v_4\}$. Summing the common odd cut parity over all K_0-components shows that the number of dirty components is even. The only component-compatible shore disjoint from either displayed dirty shore is its complement: exhaustive substitution in the preceding four equations also gives the clean pair $\{v_1, v_2\}$ and its complement, both of which intersect either dirty partition. Hence K_0 again has exactly two components.

Let k_i be the unique K_0-edge incident with v_i, and put

$$d_i = s(k_i) = c_{i-1} + c_i \neq 0.$$

On each of the two K_0-components, apply an independently chosen element of $\mathrm{GL}(2,2)$ to every low flow value. This preserves internal conservation and nonzeroness. We now choose nonzero values r_i on e_i so that

$$r_{i-1} + r_i = g_i, \tag{9}$$

where g_i is the transformed value on k_i.

For $n = 4$, the boundary values on each alternating pair agree: $d_0 = d_2 = x$ and $d_1 = d_3 = y$. Map x and y, independently, to the same nonzero α. If $\{\alpha, \beta, \gamma\} = K - \{0\}$, assigning β, γ alternately to the edges of D satisfies (9).

For $n = 5$, translate the affine colour names so $A = 0$, write $B = x, C = y, D = x + y$, and normalize $x = 1, y = 2, x + y = 3$. Thus

$$d = (3, 1, 1, 2, 1).$$

The complete repairs for the two component partitions are

K_0-components on v_i	$(g_0, \ldots, g_4)$	$(r_0, \ldots, r_4)$
$\{0,1,3\} \mid \{2,4\}$	$(3,1,2,2,2)$	$(2,3,1,3,1)$
$\{0,2,3\} \mid \{1,4\}$	$(3,3,1,2,3)$	$(1,2,3,1,2)$.

In each row the three-terminal component uses the identity. On the two-terminal component, both original boundary values are 1, which an invertible linear map sends respectively to 2 or 3. Direct cyclic substitution verifies (9).

We have constructed a nowhere-zero K-flow on every edge of G. At a cubic vertex its three incident values are therefore the three distinct nonzero elements of K, giving a Tait colouring. This contradicts the hypothesis. Thus the arbitrary extension of h was clean, and in particular h is cleanable. $\square$

Remark 3.6. The finite case split and the two repair rows are independently replayed by `verify.py` in the bundled size-five theorem package.

3.2 A line-switch lemma and the size-six/seven boundary

We need one support-preserving fact. Identify the seven nonzero full flow values with $1, \ldots, 7$ under bitwise addition, so that

$$L = \{1, 2, 3\} = \{(0, t) : t \in K - \{0\}\}, \qquad A = \{4, 5, 6, 7\} = \{(1, c) : c \in K\}.$$

Let $\mathcal{K}$ be the components of $G - h$, and let $r \in \mathbb{F}_2^{\mathcal{K}}$ indicate the rainbow-odd components. For $t \in L$, put $N_t = f^{-1}(t)$ and $P_t = \{4, 4 + t\}$. If C is a binary cycle of $G - N_t$, define

$$\tau_t(C)_W = |C \cap \delta(W) \cap f^{-1}(P_t)| \pmod 2 \qquad (W \in \mathcal{K}). \tag{10}$$

Adding t along C is a valid line-preserving switch: avoiding N_t prevents a zero value, and (10) is its change to the defect vector.

Proposition 3.7 (combined-line span)**.**

$$r \in \sum_{t \in L} \operatorname{im} \tau_t. \tag{11}$$

Consequently, if $G - h$ has at most two components, then h is cleanable by at most one valid line-preserving switch.

Proof. Suppose (11) fails. There is a vector $y \in \mathbb{F}_2^{\mathcal{K}}$ with

$$y \cdot r = 1, \qquad y \cdot \tau_t(C) = 0 \quad (t \in L,\ C \text{ a cycle of } G - N_t).$$

Let U be the union of the components selected by y, and let $u : V(G) \to \mathbb{F}_2$ indicate U. Cycle–cut orthogonality says that

$$R_t = \delta(U) \cap f^{-1}(P_t)$$

is a cut in $G - N_t$. Choose a vertex indicator x_t for that cut, and write $q = (u, x_1, x_2, x_3)$.

Define

$$\mathcal{A}(q) = \big(u(x_2 + x_3),\ u(x_1 + x_3),\ x_1x_2 + x_1x_3 + x_2x_3\big) \in \mathbb{F}_2^3.$$

Across an edge, the only possible nonzero differences of $(u; x_1x_2x_3)$, indexed by its flow value, are

$f(e)$	1	2	3	4	5	6	7
Δq	$(0; 100)$	$(0; 010)$	$(0; 001)$	$(1; 111)$	$(1; 100)$	$(1; 010)$	$(1; 001)$.

The zero difference is also allowed in every column. Direct substitution gives the pointwise identity

$$\big(\mathcal{A}(q_v) + \mathcal{A}(q_w)\big) \cdot f(vw) = \begin{cases} 1, & f(vw) = 4 \text{ and } u(v) \neq u(w), \\ 0, & \text{otherwise.} \end{cases} \tag{12}$$

Summing (12) over all edges and regrouping at vertices makes its left side

$$\sum_v \mathcal{A}(q_v) \cdot \left(\sum_{e \ni v} f(e) \right) = 0$$

by flow conservation. The right side is $|\delta(U) \cap f^{-1}(4)| = y \cdot r = 1$, a contradiction. This proves (11).

Every vector in every image of τ_t has even Hamming weight, since an affine edge between two $G - h$ components is counted at both ends. For one component the even-weight space is zero. For two components it is one-dimensional; if $r \neq 0$, (11) forces one of the three image spaces to contain its unique nonzero vector. The corresponding single valid switch kills r. A single switch has no mixed-switch quadratic correction, completing the proof. $\square$

Theorem 3.8 (minimum projections through size seven)**.** *Under the hypotheses of Theorem 3.5, every minimum extendable projection of size at most seven is cleanable.*

Proof. Only sizes six and seven remain. The four-colour circuit corollary forces h to be one circuit

$$v_0, e_0, v_1, e_1, \ldots, v_{n-1}, e_{n-1}, v_0, \qquad n \in \{6, 7\}.$$

Put $c_i = s(e_i)$, and encode the component of $G - h$ containing v_i by π_i. The c_i's form a proper cyclic word using all four affine colours. Conservation on every component W gives

$$\sum_{\pi_i = W} (c_{i-1} + c_i) = 0,$$

and its four affine cut parities are either all zero or all one.

The exact finite boundary classifier enumerates every such word and every set partition π. For each state it also enumerates one independent element of $\mathrm{GL}(2, 2)$ on the low flow values of every component and solves the n cycle equations. Its complete counts are

n	valid dirty pairs	failed linear repairs	failures with more than two components
6	2712	432	0
7	26880	5376	0

Every dirty state either produces a nowhere-zero K-flow on all of G, contradicting the non-Tait hypothesis, or has exactly two components of $G - h$. At size six the failed states may additionally be quotiented by the affine group on the four colours, the dihedral group of the cycle, and component relabelling; they give the three representatives

$(c_0, \ldots, c_5)$	$(\pi_0, \ldots, \pi_5)$
010123	001001
010232	001010
012013	000101

There are 144 labeled states per orbit. At size seven there is no residual normal form after the two-component theorem. The dependency-free enumerator regenerates all words, partitions, conservation tests, dirty states, and linear repairs rather than trusting the stored counts.

Proposition 3.7 makes every failed state clean by one valid switch. Hence no dirty size-six or size-seven global minimum is uncleanable. □

Remark 3.9. The exact classifications, proof of semantics, size-six representatives, and full standalone span proof are bundled in the cumulative through-seven package. This is a finite boundary theorem, not a search over sampled graphs.

3.3 Direct component repair through size ten

The preceding Tait repair required every new low value to be nonzero. For the original three-bit flow this is unnecessarily restrictive on h: an edge of h has first coordinate one and therefore remains nonzero even when its low value is zero. Exploiting this observation gives a stronger repair.

Proposition 3.10 (direct boundary repair certificate)**.** *Let $f = (h, s)$ be a nowhere-zero $\mathbb{F}_2 \times K$-flow. Index the vertices and edges of each circuit of h cyclically, put*

$$c_i = s(e_i), \qquad d_i = c_{i-1} + c_i,$$

and let π_i denote the component W_{π_i} of $G-h$ containing the circuit vertex v_i. Suppose there are maps $L_a \in \mathrm{GL}(2,2)$ and values $r_i \in K$ such that

$$r_{i-1} + r_i = L_{\pi_i} d_i \tag{13}$$

on every circuit, and, for every component W_a and every $c \in K$,

$$\left|\{e_i \in \delta(W_a) : r_i = c\}\right| \equiv 0 \pmod 2. \tag{14}$$

Then h is cleanable.

Proof. Apply L_a to the low value of every edge in W_a. At a vertex outside h, all three incident edges lie in the same component, so linearity preserves the flow equation. At a circuit vertex v_i, the unique edge outside h receives value $L_{\pi_i} d_i$, and (13) is exactly the missing vertex equation. Every edge outside h retains a nonzero low value because L_a is invertible. Every edge of h has full value $(1, r_i)$, which is nonzero even if $r_i = 0$. The repaired flow is therefore nowhere zero. Equation (14) is precisely the clean affine-class cut condition, component by component. □

Theorem 3.11 (minimum projections through size ten)**.** *Let G be a connected bridgeless loopless cubic graph; parallel edges are allowed. Every cardinality-minimum extendable projection of size at most ten is cleanable.*

Proof. The zero projection is clean. For a nonzero minimum projection, Corollary 3.4 says that every circuit component uses all four affine colours and hence has length at least four. The possible length partitions through total size ten are therefore

$$\begin{gathered} 4, 5, 6, 7, 8, 9, 10, \\ 4+4, \quad 4+5, \quad 4+6, \quad 5+5. \end{gathered} \tag{15}$$

For a fixed shape, the exact checker enumerates every proper cyclic word on K using all four colours on every circuit, modulo the global affine action and the exact dihedral action on each circuit (including circuit interchange when the lengths agree). For every canonical word it enumerates every set partition π of the circuit vertices. It retains exactly those partitions satisfying component conservation

$$\bigoplus_{\pi_i = a} d_i = 0$$

and the four equal affine cut parities, with at least one dirty component. It then exhausts all six choices of L_a independently for every component and all four starting values independently on every circuit; equation (13) forces the remaining r_i's. Finally it tests (14) literally.

The complete counts are

shape	canonical words	valid dirty states	failed direct repairs
4	1	1	0
5	1	2	0
6	5	27	0
7	7	116	0
8	26	1386	0
9	49	8840	0
10	151	102,290	0
$4+4$	2	114	0
$4+5$	2	372	0
$4+6$	11	7798	0
$5+5$	6	4232	0

These are all shapes in (15), and every retained boundary state has a certificate of Proposition 3.10. Hence every minimum projection in the stated range is cleanable. □

Remark 3.12. The dependency-free replay is the `verify.py` program in the bundled through-ten direct-cleaning package. It regenerates the words, partitions, conservation tests, and repairs. A separately written through-nine package agrees on the overlapping range, and two independent agent runs reproduced the size-ten counts. These checks are not independent human peer review.

3.4 The first direct-repair failures and size eleven

The direct-cleaning statement in Theorem 3.11 is not universal. A second elementary certificate explains why its first failures nevertheless cannot be globally minimum.

Proposition 3.13 (circuit-deletion certificate). *Let $f = (h, s)$ be a nowhere-zero $\mathbb{F}_2 \times K$-flow, and let D be a circuit component of h. Suppose component maps L_a and values r_i satisfy the vertex equations (13), and suppose $r_i \neq 0$ on every edge of D. Then $h - D$ is extendable.*

Proof. Use first coordinate one on $h - D$ and zero on D. On all circuit edges use the low values r_i, and inside each component of $G - h$ use $L_a s$. The proof of Proposition 3.10 verifies all vertex equations and nonzeroness off D. On D, the first coordinate is now zero but the assumed low value is nonzero. The result is therefore a nowhere-zero flow with first-coordinate support $h - D$. □

Theorem 3.14 (minimum projections through size eleven). *Under the hypotheses of Theorem 3.11, every cardinality-minimum extendable projection of size at most eleven is cleanable.*

Proof. Only total size eleven remains. The four-colour circuit condition gives the three shapes

$$11, \qquad 4+7, \qquad 5+6. \tag{16}$$

The same exact boundary enumeration used above gives

shape	canonical words	charge-valid	dirty	failed direct repairs
11	345	1,183,710	919,764	0
$4+7$	17	57,500	46,786	0
$5+6$	26	87,884	71,170	14

The 14 literal failures form 11 orbits under the full affine, dihedral, and component-renaming action. For each failure, the checker exhausts the same component maps and independent circuit starts and finds a certificate satisfying (13) in which every low value on the six-circuit is nonzero. Proposition 3.13 then produces an extendable projection supported only on the five-circuit. Its size is five, contradicting the assumed global minimality of the size-eleven projection. Hence every globally minimum size-eleven state is directly cleanable. □

Remark 3.15 (sharp direct-repair countermodel). The 14 failed boundary states are realizable. Replacing each two-vertex component block by an edge and each three-vertex block by a claw yields, in all 14 cases, the same simple bridgeless non-Tait graph on 12 vertices, with graph6 encoding

`Kt?G?DIPOqCo`.

It is the triangle-expanded Petersen graph. Exact cycle-space enumeration shows that the displayed size-eleven 5+6 projection is extendable but uncleanable, while the minimum extendable-projection size is five and all six size-five minima are cleanable. Thus it refutes the direct-repair strengthening, not FiveCDC or the minimum-projection conjecture. The cumulative checker and all 14 certificates are in the bundled through-eleven package.

3.5 Clean or delete at size twelve

Propositions 3.10 and 3.13 give a useful dichotomy: it is enough to find component maps and circuit starts that either make every affine cut parity even, or make one whole support circuit low-valued and nowhere zero. The first branch cleans the projection; the second strictly shortens it and is therefore impossible for a global minimum.

Theorem 3.16 (minimum projections through size twelve)**.** *Under the hypotheses of Theorem 3.11, every cardinality-minimum extendable projection of size at most twelve is cleanable.*

Proof. Only total size twelve remains. The complete circuit-length list is

$$12, \qquad 4+8, \qquad 5+7, \qquad 6+6, \qquad 4+4+4. \tag{17}$$

For each shape, the exact classifier enumerates the same word orbits and charge-valid component partitions as before. It first exhausts every component map and relative circuit start for direct cleanliness. If that fails, it tests whether the integrated word on a circuit omits a colour. Translating that circuit by the omitted colour makes all of its low values nonzero, giving the hypothesis of Proposition 3.13. The mutually exclusive counts are

shape	words	dirty	direct clean	delete
12	1014	11,465,978	11,465,978	0
$4+8$	66	773,721	773,717	4
$5+7$	64	738,969	738,763	206
$6+6$	66	774,196	774,014	182
$4+4+4$	3	35,960	35,960	0

There are zero states in neither branch. The 392 deletion states have literal component maps, integrated circuit words, and omitted-colour certificates in the frozen output. Since a globally minimum projection cannot take the deletion branch, every size-twelve minimum is cleanable. □

Remark 3.17. The primary C++ classifier and certificate parser are bundled in the through-twelve clean-or-delete package. A separately written C++ exact cover auditor independently regenerates the word orbits, valid component blocks, map search, clean decisions, and all 392 deletion decisions. The two implementations agree on every table entry. This is independent agent replication, not human peer review.

Corollary 3.18 (rainbow shortcut)**.** *Every binary cycle C satisfying $|C \cap h| > |C - h|$ meets all four sets M_c.*

Proof. If C omitted one class M_c, it would contradict (6). □

4 The exact missing step

Theorem 3.1 suggests the following selection principle.

Conjecture 4.1 (minimum extendable projection)**.** Every bridgeless cubic graph has a minimum-cardinality extendable projection that is cleanable.

The stronger statement that every minimum projection is cleanable also survives the finite census in Section 5, but it is not needed. Conjecture 4.1, together with the Hušek–Šámal equivalence, would imply the cubic FiveCDC conjecture.

Suppose a minimum projection h is not cleanable. Every extension then has a rainbow-odd component W of $G-h$, so all four M_c's cross $\delta(W)$ oddly. A direct proof of Conjecture 4.1 would need to derive either

1. a binary cycle C avoiding some M_c and satisfying $|C \cap h| > |C - h|$, contradicting Theorem 3.1; or
2. a sequence of support-neutral low-coordinate recolourings followed by such a strict exchange.

The second possibility cannot currently be discarded. Small exact examples have four disjoint colour classes meeting every immediately component-reducing cycle, while a support-preserving recolouring escapes the trap. Other examples show that a fixed nowhere-zero Fano flow can have all seven of its functional projections dirty. These are proof-strategy obstructions, not FiveCDC counterexamples. They rule out an argument that fixes an arbitrary initial flow or assumes one elementary switch always suffices.

The unresolved human proof obligation is precise: from a rainbow-odd component of an *uncleanable globally minimum-support projection*, derive a negative colour-avoiding cycle after allowable neutral recolouring, using the four simultaneous shortest-T_c-join inequalities of Corollary 3.2. No proof of this statement is given here. By Theorem 3.16, any counterexample to this selection principle has minimum support at least thirteen.

5 Exact finite census

5.1 Canonical order-18 method

The accompanying checker uses canonical generation rather than sampling. For order 18, the command

```
geng -cq -d3 -D3 18
```

generates every connected simple cubic graph once. The checker then:

1. tests bridgelessness by deleting every edge and checking connectivity;
2. tests Tait colourability by exhaustive backtracking;
3. enumerates the entire binary cycle space;
4. processes nonzero projections h by increasing cardinality;
5. quotients pairs of cycles (p, q) by their exact semantic state $(p \cup q, p \cap q)$;
6. tests extendability by $E - h \subseteq p \cup q$; and
7. computes the components of $G - h$ and tests (5) on every component.

For a connected cubic graph of order 18, the binary cycle-space dimension is

$$|E| - |V| + 1 = 27 - 18 + 1 = 10.$$

Thus all $2^{10} = 1024$ binary cycles are enumerated. Equal union/intersection pairs have exactly the same coverage and component parity behaviour, so the semantic quotient loses no information. No snark or cyclic-connectivity reduction is used.

class	count
connected simple cubic graphs	41,301
bridgeless graphs	39,866
bridgeless non-Tait graphs	179

Table 1: Complete order-18 population.

5.2 Results and reproducibility

Every minimum-cardinality extendable projection of every one of the 179 non-Tait graphs is cleanable. The minimum-size histogram is

$$5 : 166, \qquad 6 : 11, \qquad 7 : 1, \qquad 8 : 1.$$

The checker hashes, in canonical graph order, the graph6 string, minimum size, number of minimum extendable projections, and number of cleanable minimum projections. The resulting SHA-256 is

`82b01cc2f01d4f50f7745f4bdb1b3dc46ade798d452a9d4f11d4a91d738a7834`.

From the project root, run:

```
python3 scratch/check_husek_samal_minimum_projection_frontier.py \
  --order 18
```

The checker requires Python, the sibling cycle-space module `scratch/search_fano_all_bad_projection_subspaces.py`, and nauty `geng` on `PATH`. It has no SAT solver, network, or non-standard Python-package dependency. The relevant source hashes are

frontier checker:
`37e972782863c7a9506f6d339159ddaca3865363107704f86365c1a9b77a30c3`
cycle-space module:
`7e5ea2599e7533f041ea88de5c6f5c7a3bf542cedc33d78728439d43ce8861e8`

This is a finite theorem. It is evidence for Conjecture 4.1, not a proof of it.

5.3 Expanded frozen-source replay

A separate C++ classifier implements the same extendability and cleanability semantics by Gaussian elimination over $\mathbb{F}_2$. It replays two previously frozen graph6 populations:

1. 14,009 cyclically 4-edge-connected non-Tait records of even orders 10 through 28; and
2. 12,892 bridgeless non-Tait rows from a broader hard order-22 source.

The exact results are:

For the cyclically 4 source the minimum-size profile is

$$5 : 14007, \qquad 6 : 1, \qquad 9 : 1.$$

For the hard order-22 source it is

$$5 : 11776, \quad 6 : 1025, \quad 7 : 55, \quad 8 : 20, \quad 9 : 6, \quad 10 : 10.$$

These counts refer to graphs; the third column of the table counts all minimum projections on those graphs.

From the project root, run:

frozen source	graphs	minimum projections	uncleanable
cyclically 4, orders 10–28	14,009	92,754	0
hard order 22	12,892	54,785	0
total	26,901	147,539	0

Table 2: Exact minimum-extendable-projection replay on frozen sources.

```
cd scratch/minimum-projection-census-through28-20260729
python3 verify_summary.py
```

The replay compiles the classifier, verifies every input digest and every emitted graph6 row, and checks all aggregates above. Its checksum ledger is the `SHA256SUMS` file in that package. The unconditional claim here concerns the literal frozen files. Their identification with complete graph classes uses the documented upstream canonical-generation and filtering packages; this replay does not independently regenerate those populations.

5.4 Retained strong snarks of orders 34 and 40

A separate exact scan covers the literal retained strong-snark files at orders 34 and 40: Every row

order	graph6 rows	minimum projections	uncleanable
34	7	371	0
40	7,654	433,359	0
total	7,661	433,730	0

Table 3: Exact minimum-projection scan on the frozen strong-snark rows.

has minimum extendable-projection size ten. An ordinary-CNF solver encodes the two low binary cycles, coverage on $E - h$, product variables for $p_e \wedge q_e$, and an XOR chain imposing even product parity on every component cut. Every even projection through weight twelve is enumerated as a vertex-disjoint union of circuits, which is exhaustive for a cubic graph. All rows reach their first extendable level at weight ten.

The order-34 output and the first five order-40 rows were reproduced independently by the linear-algebra scanner used in the earlier census. The full order-40 claim uses the exact SAT implementation. The concatenated result digest is

`becb7e4b648c000ec874509bccdf389df24f6f0d82bce2da49ba10d5b10d38e3`.

Inputs, frozen output, both scanner implementations, and the replay script are under `scratch/`, in the subdirectory `minimum-projection-known-strong-snarks-20260729/`. As before, the unconditional statement concerns the literal bundled rows; completeness of the named graph populations is inherited from upstream provenance.

5.5 A certified 130-vertex parameter-separation test

The retained 130-vertex exchange graph gives a substantially larger exact stress test. A three-coordinate CNF uses binary-cycle variables (h_e, p_e, q_e), the three parity equations at every vertex, the nowhere-zero clauses $h_e \vee p_e \vee q_e$, and a sequential cardinality counter. Its certified results are

$$\min |h| = 42, \qquad \#\{h : |h| = 42,\ h \text{ extendable}\} = 11264,$$

and every one of those 11,264 supports is cleanable. The size-$\leq$ 41 CNF has 7,760 variables and 16,218 clauses. A second CNF blocks every listed size-42 support and proves enumeration completeness. Both UNSAT LRATs are accepted by C `lrat-check` and verified CakeML `cake_lpr`; every positive clean lift is also checked directly.

On the same graph the different parameter

$$\rho_3(G) = \min_{f,\ a \neq 0} |f^{-1}(a)|$$

equals five, while no size-five value class packs two boundary joins. Thus minimizing one nonzero Fano value class and minimizing an entire coordinate projection are not interchangeable. The former selection principle fails on this graph; the latter passes all 11,264 minimum cases. To replay the package, run:

```
cd search/minimum-projection-n130-20260729
python3 verify.py
```

6 Discussion

The exchange theorem converts global support minimality into four ordinary geodesic conditions: one shortest join for each affine value class. Theorems 3.5, 3.8, 3.11, 3.14, and 3.16 close every possible nonzero support size through twelve: a counterexample to Conjecture 4.1 must have minimum projection size at least thirteen. The useful feature is simultaneity. A cycle that would shorten the projection is allowed to be blocked by one colour, but Corollary 3.18 forces it to be blocked by all four. Likewise, disconnected projections are severely constrained because every circuit component must contain every colour.

The four-colour simultaneity cannot be replaced by a theorem about one matching. In the Petersen graph with edge set

$$\{03, 06, 09, 14, 16, 18, 25, 26, 27, 37, 38, 47, 49, 58, 59\},$$

the matching $M = \{06, 14, 25\}$ is contained in exactly two minimum-cardinality binary cycles:

$$\{06, 09, 14, 18, 25, 26, 49, 58\}, \qquad \{06, 09, 14, 16, 25, 27, 47, 59\}.$$

Both have size eight, and in both cases the components of the complement have M-endpoint parities $1, 0, 1$. Exhausting all 2^{15} edge subsets proves minimality. The complete finite replay is in `scratch/petersen-minimum-tjoin-countermodel-20260729/`. This obstruction does not realize four synchronized affine classes.

What is not yet understood is how the common rainbow-odd cut interacts with the four shortest-join duals. A publishable resolution would need either a cut/cycle contradiction, a neutral-reconfiguration theorem, or an exact counterexample to Conjecture 4.1. Until one of those is supplied, the correct status is a frontier result.

AI-use and verification disclosure

This research route, proof draft, checker, computation, and exposition were developed by OpenAI Codex agents under Atharva Vaidya's direction. The minimum-projection exchange proof, the size-at-most-five proof, and the span proof used at sizes six and seven are displayed in full so that they can be checked without trusting an AI system. Those boundary classifications and the stronger direct-cleaning theorem through size ten are also regenerated by dependency-free exact checkers. The through-eleven checker freezes every first direct-repair failure and its smaller-support certificate. The through-twelve package freezes all 392 new circuit-deletion certificates and includes a separately implemented full census. The order-18 result is reproducible from canonical generation and exhaustive cycle-space enumeration. The expanded census exactly replays frozen sources but inherits their population provenance. Both remain computational results and should be independently rerun.

Hušek and Šámal's flow criterion and Hoffmann-Ostenhof's matching/join formulation are prior work. This draft makes no claim of a FiveCDC resolution and no priority claim for the minimum-projection viewpoint pending expert literature review. Before public submission, a human graph theorist should check the proof line by line, rerun the census in a clean environment, audit the bibliography, and decide authorship and venue policy under the applicable AI-disclosure rules.

References

[1] Uldis Alfred Celmins. *On Cubic Graphs That Do Not Have an Edge-3-Colouring.* PhD thesis, University of Waterloo, 1985.

[2] Arthur Hoffmann-Ostenhof. A note on 5-cycle double covers. *Graphs and Combinatorics*, 29:977–979, 2013.

[3] Radek Hušek and Robert Šámal. Exponentially many circuit double covers, 2026. Version 1, submitted July 27, 2026.

[4] Myriam Preissmann. *Sur les colorations des arêtes des graphes cubiques.* PhD thesis, Université de Grenoble, 1981.